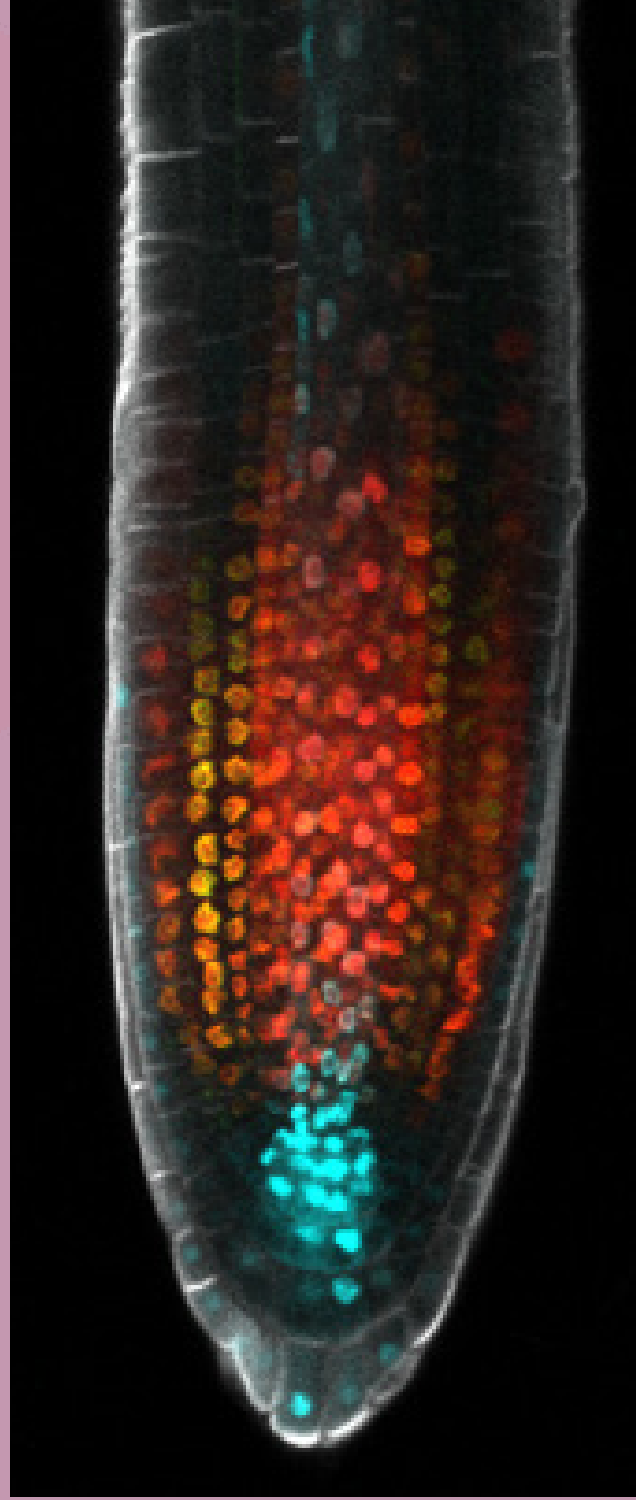




PhD candidate Gyöngyi Macías Honti

I am interested in knowing what happens in the root at a cellular level during flooding, what the cells experience, and how they communicate with each other in these conditions. I study young *Arabidopsis* plant roots.



Modified *Arabidopsis* root under microscope. Yellow-to-red circles signals (called R2D2) indicate a ratio of plant hormone auxin, and blue signal indicates molecular responses to auxin.

Modified *Arabidopsis* root under microscope. R2D2 yellow-to-red circles as well. Green signal shows localization of PIN1 protein, which transports auxin in the vasculature of the roots.



ON THE IMPACT

The first step is always to understand what is happening to the plant so we can eventually apply the knowledge into a broader perspective for better food production.

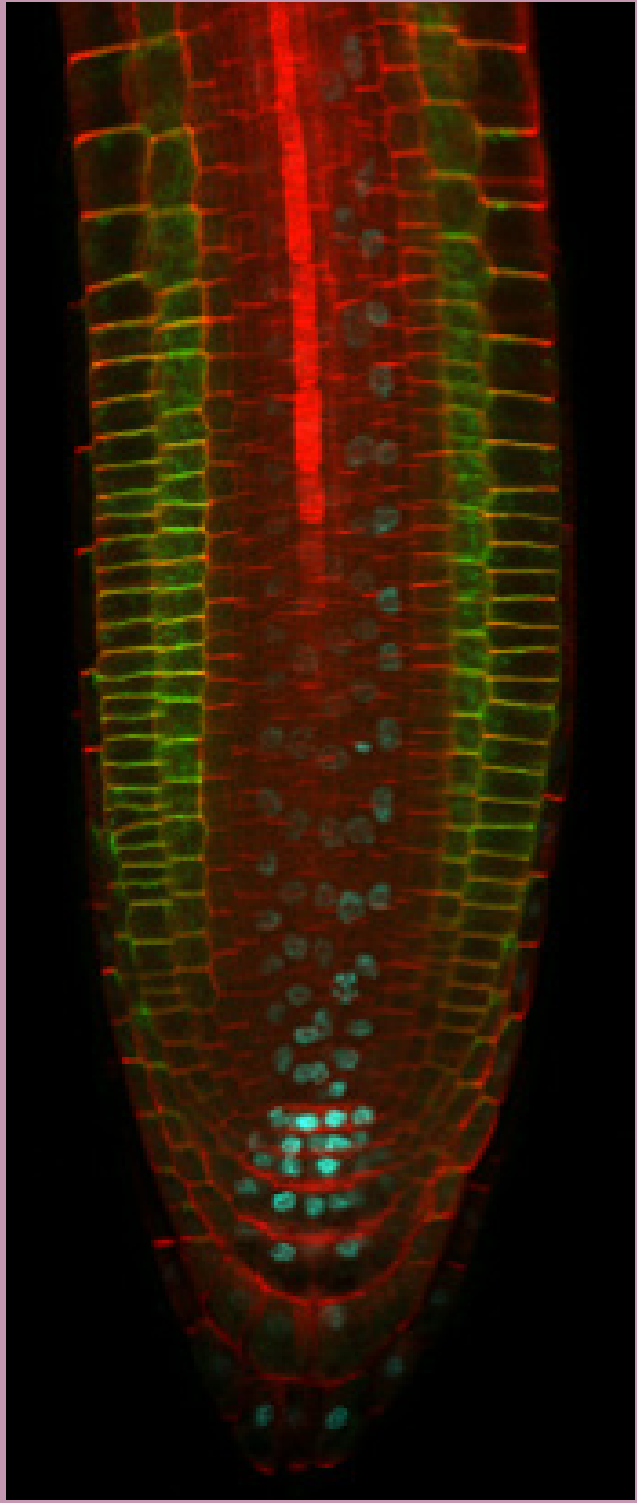


Normal *Arabidopsis* root under microscope. It is stained with Direct red 23 to show the cell walls of each cell in the root.

ON PREVIOUS EXPERIENCES

During my bachelor's, I taught Biology to middle school kids in low-income areas of Mexico. That was an enriching experience because both the kids and their parents were very enthusiastic and left the workshops more curious about science.

Modified *Arabidopsis* root under microscope. Green signal is the PIN2 protein which transports auxin, and blue dots indicate auxin response.



ON THE CHALLENGES

When we explain our research to the general public, we often oversimplify it. This is very detrimental to us, because we don't allow people to understand how we create value to society with our research. If everybody hides behind this same script and the same narrative, the public may see us as unreachable.

It's in our benefit for people to understand the research to help them see the purpose of it. We should find a common language that doesn't oversimplify the research, while still being accessible to those without specific knowledge in the field. Let both sides put in the effort to understand each other without fear of rejection, and surprise ourselves with what eventually comes on the other side.

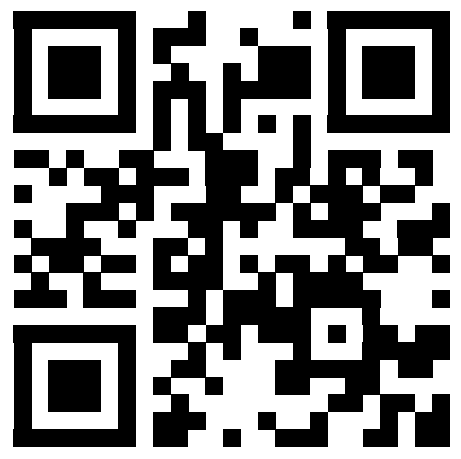
ON THE FUTURE

We think there are a few things we can do as scientists for science communication:

Gyöngyi - Have an activity where we can reach the elderly. In turn, they can share what they learn with their families and friends as well.

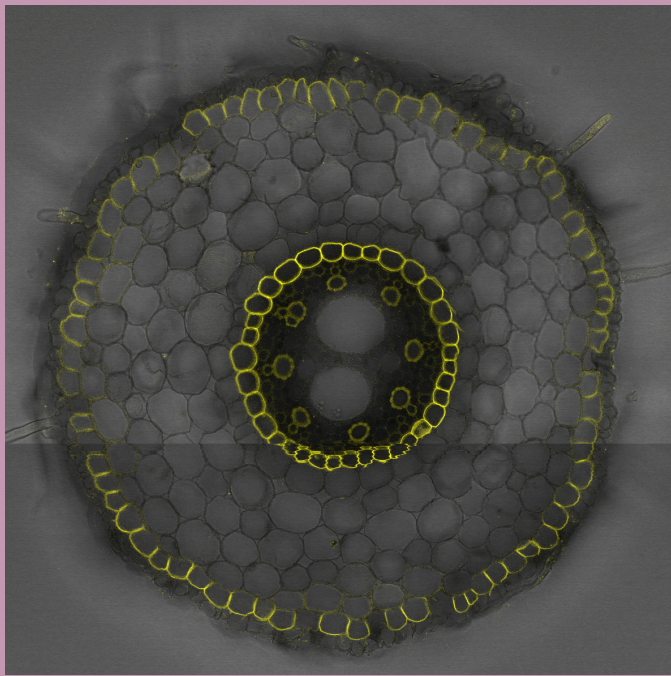
Gyöngyi - Change the way we treat science communication within research groups by making low-threshold activities mandatory. For example, a group blog or social media page where all group members have to post in rotation once every few weeks.

Read this poster in Dutch

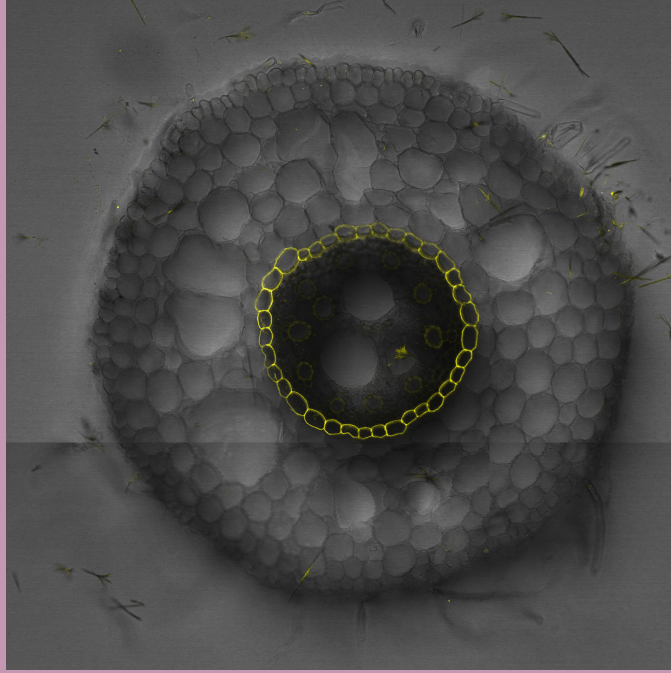


PhD candidate Julia Mars

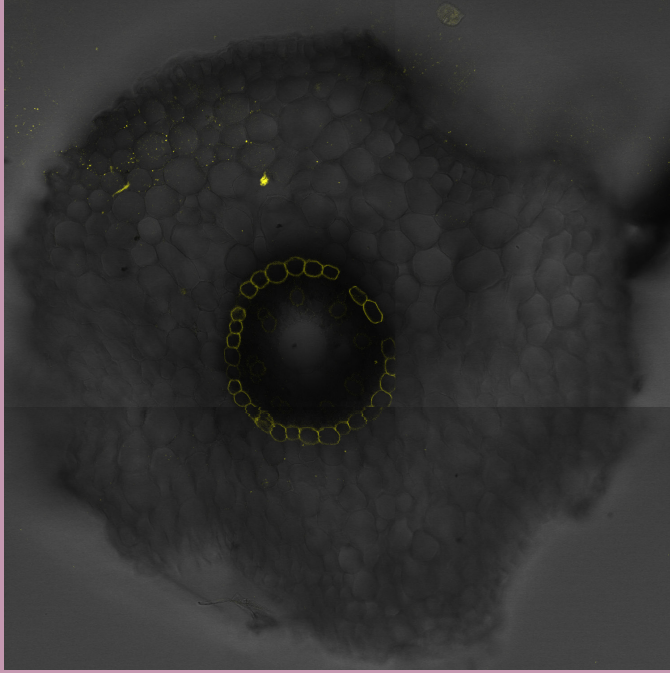
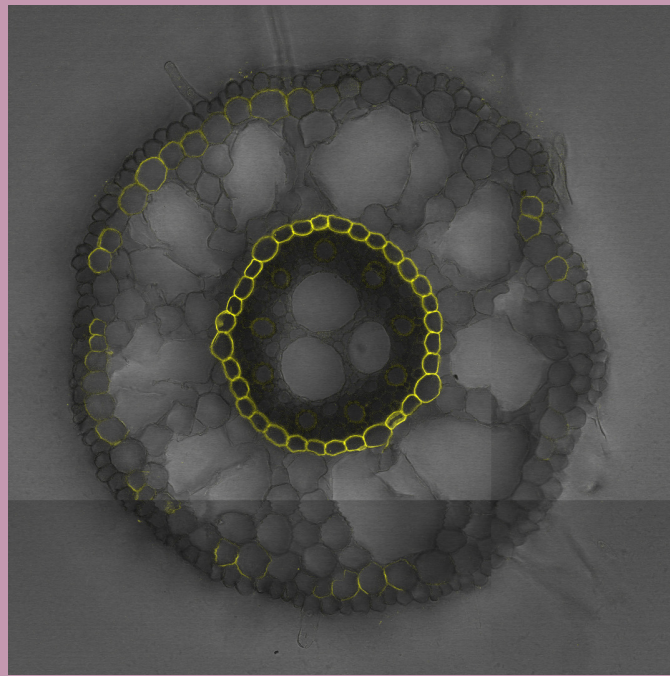
Like our intestinal microbiome, plants have their own microbiome in the soil surrounding the roots. I'm investigating how this microbiome affects plant's health, focusing on sorghum. Specifically, how these bacteria can enhance the production of suberin (cork) in plant roots. Cork helps keep water in or out and has other protection roles within plant roots.



Microscope images of sorghum root cross sections. Wherever there is yellow, it means cork is present. These two correspond to the 100% and 50% of the length of the root.



Microscope images of sorghum root cross sections. Wherever there is yellow, it means cork is present. These two correspond to the 75% and 25% of the length of the root.



ON PREVIOUS EXPERIENCES

During my research, I get a lot of help from students in the lab. I really enjoy guiding them through their projects and thinking with them about what the next scientific questions could be, and how to make decisions based on their results. I think spreading my knowledge in this way can already have quite an impact. It starts with a small circle of people, but from there on it spreads.

ON THE IMPACT

Many scientific findings are now considered general knowledge. Take photosynthesis for example. This process is now something that people are aware of even though you cannot see it happening. Without plant research, we would never have known. In that sense, I think scientists have a great societal contribution.

ON THE CHALLENGES

In my experience, most people are really interested in what is happening in science. However, sometimes they think it could be "too hard" to understand science. I think it's more about just giving people the tools and the language to think with you instead of us lecturing them.

Nowadays, it seems like the most valid argument to justify importance is to make something relevant through economic gain. Not every part of science and research should be seen through this lens. However, it feels taboo to say that you just want to explore a certain scientific field for the sake of generating knowledge.



Young sorghum root after staining. The difference in brightness helps us understand how sorghum uses cork.

Julia - Take people along in the explanation, step by step, using common words. Otherwise, it can be too overwhelming.

Julia - Start an explanation with a question that tingles the mind and can get people interested and excited to know more.

What do you think?

